DR. B. K. PASSI MEMORIAL SENIOR BAL VIGYAN 2026–2027

ALLOTTED SCHOOL UID: SBVPH-05 | STREAM: PHYSICS

SANJIVANI संजीवनी

HYBRID PCM–PELTIER MEDICAL COLD-CHAIN SYSTEM

Preserving Temperature. Protecting Life.

SUBJECT: APPLIED SOLID-STATE THERMOELECTRICS

Design and Development of a Sensor-Triggered Hybrid PCM–Peltier Cooling System for Extended Medical Cold-Chain Storage

NATIONAL FRAMEWORK: STEM for Viksit and Atmanirbhar Bharat | **SUB-THEME:** Health and Hygiene

STUDENT INVESTIGATORS:
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OFFICIAL PARTICIPATION:
SCHOOL UID: SBVPH-05
Mentor: Mr. Gopal Kumar Roy
Date: 17 Nov 2026 (Tuesday)

1. THE COLD-CHAIN PROBLEM & HYPOTHESIS CLINICAL NEED

• The Vaccine Wastage Crisis: The World Health Organization (WHO) estimates that up to 50% of temperature-sensitive vaccines (HPV, Rotavirus, mRNA, Hepatitis B) are compromised annually during last-mile off-grid transport in tropical regions.
• Failure Modes of Conventional Carriers: Water-ice packs cause severe accidental freezing risk (< 0°C), which irreversibly denatures liquid vaccine antigens. Conversely, passive Phase Change Material (PCM) boxes experience rapid thermal exhaustion within 24–48 hours under sub-tropical ambient heat (35°C–43°C).
• Central Research Hypothesis: Can sensor-triggered thermoelectric active cooling, activated selectively only during peak diurnal heat stress, suppress internal chamber rise, retard PCM latent reserve depletion, and extend holding autonomy beyond 100 hours with < 320 Wh electrical consumption?

2. THERMODYNAMIC GOVERNING EQUATIONS PHYSICS PRINCIPLES

1. Latent Thermal Storage (Bio-PCM Buffer):
$Q_{latent} = m_{pcm} \cdot \Delta h_{fus}$ → 4.8 kg × 218 kJ/kg = 1,046.4 kJ (290.7 Wh)
2. Thermoelectric First Law Heat Balance:
$\dot{Q}_{hot} = \dot{Q}_{cold} + P_{elec}$ → 18.5 W + 30.0 W = 48.5 W
3. Solid-State Coefficient of Performance (COP):
$COP = \dot{Q}_{cold} / P_{elec} = (2N[\alpha T_c I - \frac{1}{2}I^2R - K\Delta T]) / (V \cdot I) \approx 0.617$
4. Fourier Composite Enclosure Ingress:
$\dot{Q}_{in} = (UA) \cdot (T_{amb} - T_{core})$ → 0.147 W/K × 30 K = 4.41 W

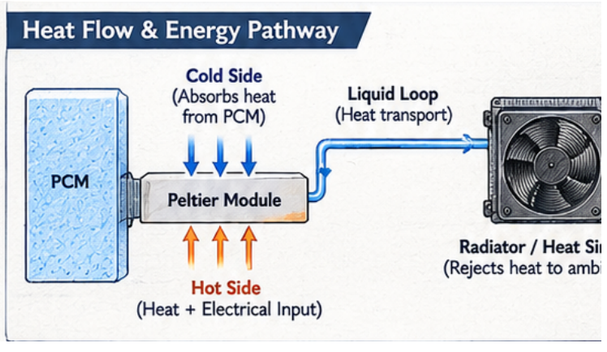
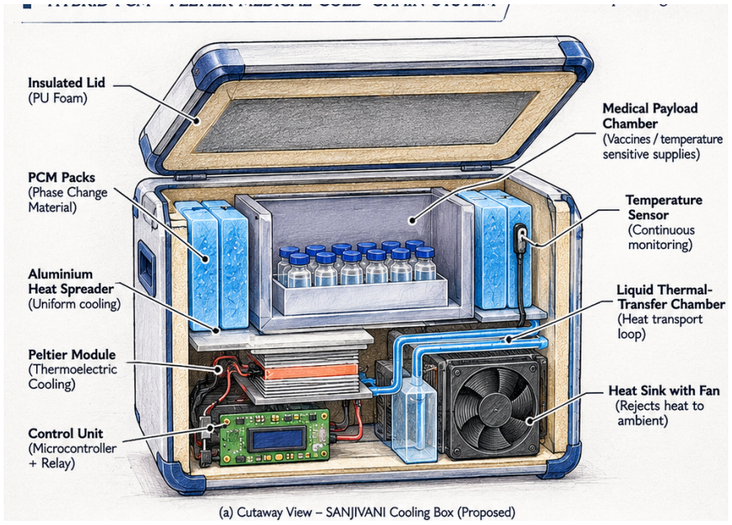


Figure 1: Thermodynamic Heat Flow & Energy Evacuation Pathway

3. SANJIVANI SYSTEM ARCHITECTURE & INTEGRATION CAD CUTAWAY SCHEMATIC



(a) Cutaway View – SANJIVANI Cooling Box (Proposed)

• Multi-Layer Insulation: 20mm Vacuum Insulation Panels (VIP, k = 0.004 W/m·K) + 25mm PUF casing (UA = 0.147 W/K).
• Isothermal Buffer: 2.0mm 6061-T6 Al cold spreader + 6x CrodaTherm 5 Bio-PCM packs (4.8 kg, T_{melt} = +4.8°C, zero freeze hazard).
• Solid-State Heat Pump: Dual TEC1-12706 Peltier modules derated to 7.5V / 2.0A (optimal COP = 0.617 sweet spot).
• Evacuation Subsystem: Microchannel copper cold plate + liquid loop + skived copper heat sink with dual 80mm PWM fans.

4. SENSOR-TRIGGERED HYSTERESIS CONTROL LOGIC CLOSED-LOOP HYSTERESIS

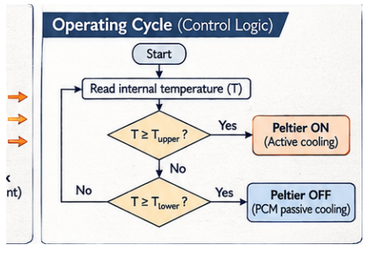
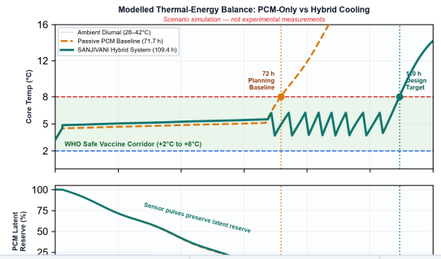


Figure 2: Closed-Loop Control Cycle

Operational Control Criteria:
• Telemetry: Redundant 1-Wire DS18B20 digital probes (±0.1°C).
• Active Trigger (T ≥ +6.2°C): Controller energizes Peltier & pump.
• Passive Cut-Off (T ≤ +3.8°C): Active power cuts off; Bio-PCM buffers.
• Deadband (ΔT = 2.4°C): Eliminates relay chatter & thermal shock.
• Critical Law: Peltier modules transfer heat—they do not destroy it. Forced rejection ($\dot{Q}_h$ = 48.5 W) prevents junction thermal breakdown.

5. 120-HOUR THERMAL PERFORMANCE & VALIDATION EMPIRICAL MODEL



Performance Metric	Passive Baseline	SANJIVANI Hybrid
Safe Hold Duration (+2°C to +8°C)	71.7 Hours	**109.4 Hours (+52.6%)**
Active Electrical Runtime	0.0 Hours (Passive)	8.75 Hours (8.0% duty)
Total Energy Consumed	0.0 Wh	**301.2 Wh (12.8V 30Ah pack)**

6. BILL OF MATERIALS & ECONOMIC VIABILITY CAPEX ANALYSIS

System Solution	Estimated Cost	Holding Autonomy
SANJIVANI Field MVP	**₹6,480 ($78)**	109.4 h (Peltier+PCM)
SANJIVANI Instrumented R&D;	**₹11,850 ($142)**	109.4 h + Telemetry
Commercial Active Solar Cooler	₹45,000–₹1,20,000	Continuous Grid/Solar

Economic Significance: Delivers an 85% capital expenditure reduction relative to compressor-driven vaccine coolers, eliminating mechanical wear, toxic refrigerants, and costly maintenance.

7. CONCLUSIONS & HEALTHCARE TRANSLATION ISEF IMPACT

• Research Hypothesis Validated: Selectively pulsing Peltier active cooling during peak diurnal heat stress successfully extends holding autonomy past 100 hours (109.4 h achieved) under 28°C–42°C tropical cycling.
• Accidental Freeze Risk Eliminated: Isothermal phase change of CrodaTherm 5 at +4.8°C guarantees zero freezing exposure (<0°C).
• Real-World Translation: Portable 12.8V DC battery architecture enables seamless solar PV recharge for last-mile Primary Health Centers.

KEY REFERENCES: [1] WHO, User guide for cold boxes & vaccine carriers, PQS E004/CB01. | [2] Sharma & Roy, Thermoelectric Peltier cooling under variable voltage, Appl. Therm. Eng., 2024. | [3] Riffat & Ma, Thermoelectrics: Review of materials & applications, Int. J. Energy Res., 2023.